Below is a summary comparison of the datasets for Boston, NYC, and Chicago, focusing on their strengths and limitations for a project analyzing mobility patterns and air quality:

Boston

- **Data Provided:**
  – *Boston_PM_NO2.csv* (focused on NO_2 levels)
  – *Boston.geojson* (spatial boundaries for mapping)
- **Benefits:**
  – **Simplicity & Focus:** Offers a straightforward dataset centered on a key pollutant (NO_2), making it easy to map and analyze.
  – **Clean Spatial Context:** The geojson file aids in visualizing spatial distributions of air quality.
- **Disadvantages:**
  – **Limited Scope:** Covers only NO_2, lacking multi-pollutant insights.
  – **No Mobility Data:** Essential mobility metrics are absent, which limits the analysis of how movement impacts air quality.
  – **Single-City Perspective:** Without additional data, it only represents Boston.

NYC

- **Data Provided:**
  – *NYC_PM_Part1.csv, NYC_AQ.csv, NYC_PM.csv* (comprehensive air quality measurements)
  – *nyc_polygon.geojson* (for spatial mapping)
  – *nyc1.csv* (potentially includes mobility data or additional metrics)
- **Benefits:**
  – **Comprehensive Air Quality Coverage:** Multiple files allow analysis of various pollutants and aspects of air quality.
  – **Potential Mobility Insights:** If *nyc1.csv* contains mobility metrics, it directly supports the project's objective of linking mobility patterns with air quality.
  – **Enhanced Spatial Analysis:** The geojson data enables detailed heatmaps and geographical visualizations.
- **Disadvantages:**
  – **Complexity:** Integrating multiple datasets may require significant data cleaning

and consolidation efforts.
– **Uncertainty:** The exact content of *nyc1.csv* is ambiguous; if it lacks mobility data, additional mobility datasets will be necessary.

Chicago

- **Data Provided:**
  – *chicago_eclipse_data_part_1.csv*
- **Benefits:**
  – **Event-Specific Analysis:** Captures data during an eclipse event, which could reveal how such anomalies affect mobility and air quality.
  – **Integrated Opportunity:** May include both mobility and air quality data if designed to study the event's impact.
- **Disadvantages:**
  – **Limited Representativeness:** Being event-specific, it may not reflect typical patterns, making cross-city comparisons challenging.
  – **Variable Uncertainty:** It is unclear whether the dataset includes comprehensive mobility metrics.
  – **Comparability Issues:** The event-driven nature might not align well with routine datasets from Boston or NYC.

Overall Recommendation

- **Best for Comprehensive Analysis:**
  NYC Data appears to be the most promising choice because it offers a richer set of air quality data and has the potential for mobility insights (pending confirmation of *nyc1.csv* content). This makes it more adaptable for clustering, regression, and spatial visualizations.
- **Considerations:**
  – **Boston Data** is easier to manage but will require additional mobility data to fully address the project's aims.
  – **Chicago Data** is excellent for a focused case study on event-driven impacts, but its specialized nature may limit its use in a broad comparative analysis across cities.